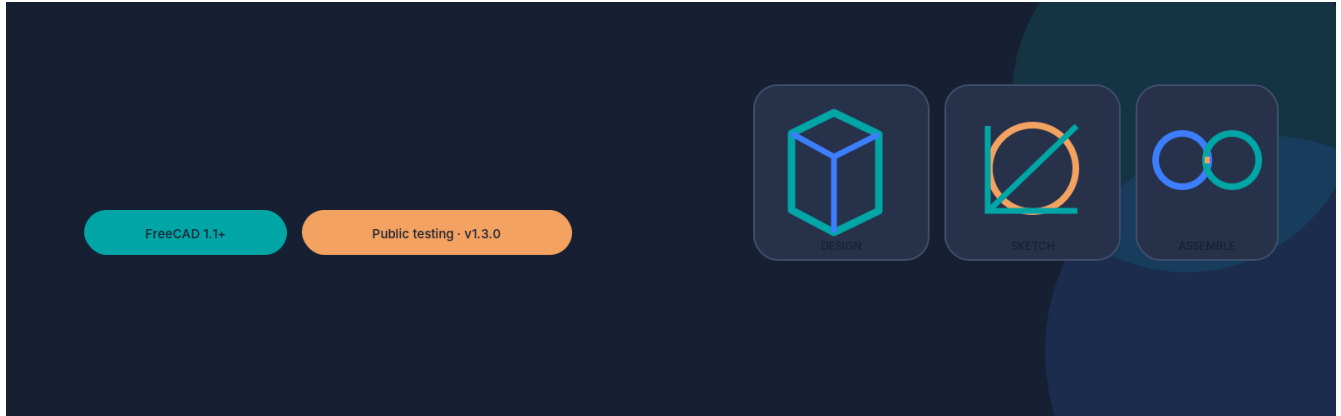


Fusion-like FreeCAD Workflow Profile v1.3.0

Graphical user, installation, and public-testing guide

Version 1.3.0

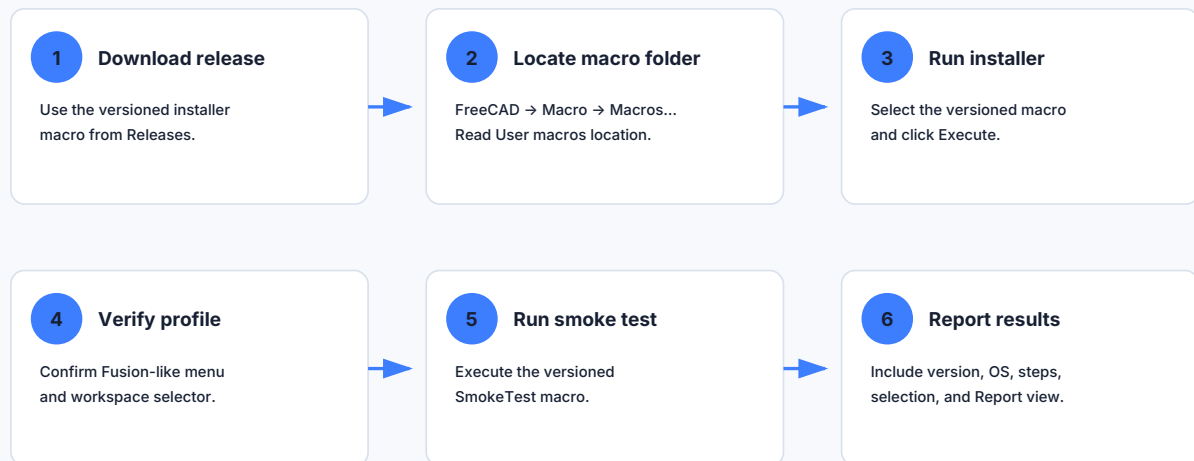


Independent public-test project for FreeCAD 1.1+. No Autodesk assets. Not affiliated with Autodesk or the FreeCAD project.

Installation and setup guide

Install and verify

Recommended public-test setup



Requirements

- FreeCAD **1.1 or newer** with the standard Part Design, Sketcher, TechDraw, and Assembly modules available.
- Permission to write to your FreeCAD user-data directory.
- A backup of important .FCStd documents before public-testing a new build.

The installer checks the FreeCAD version before writing the add-on. Use FreeCAD's own **Macro → Macros...** dialog to locate the macro directory; this is more reliable than hard-coded operating-system paths.

Recommended installation

1. Close active Sketch, Hole, Thread, Drawing, or Assembly task dialogs.
2. Download macros/Install_FusionLike_FreeCAD_v1.3.0.FCMacro.
3. Open FreeCAD.
4. Choose **Macro → Macros...**
5. Read the **User macros location** shown in the dialog.
6. Copy the installer macro into that directory.
7. Return to the Macro dialog, select the installer, and click **Execute**.
8. Confirm the installation message.
9. Verify that a **Fusion-like** menu appears and that the top workspace selector contains **DESIGN**, **ASSEMBLE**, and **DRAWING** when those native workbenches are present.

Restarting is normally unnecessary. Restart once when upgrading from a runtime that has been loaded for a long session, or when an old native task dialog remains open after installation.

What the installer writes

The installer creates this user module:

```
<FreeCAD user data>/Mod/FusionLikeUI/  
├─ Init.py  
├─ InitGui.py  
├─ fusion_like_ui_runtime.py  
└─ README.txt
```

InitGui.py waits for the FreeCAD main window, imports the runtime, and retries startup if the GUI is not ready yet.

The installer also stores profile state under:

```
User parameter:BaseApp/Preferences/FusionLike
```

That state includes whether the profile is enabled, the installed version, the original workbench/layout state, and selected workflow preferences.

Upgrade from v1.0-v1.2

Run the v1.3 installer directly over the earlier version.

- Do **not** run the uninstaller first.
- The original interface backup captured during the first installation is preserved.
- The loaded older runtime is asked to restore and remove its event hooks before replacement.
- Existing modeled documents are not rewritten by the installer.

Verify the installation

Run macros/FusionLikeUI_SmokeTest_v1.3.0.FCMacro through **Macro → Macros...**

The smoke test checks:

- installed version and runtime path
- required public runtime functions
- availability of key FreeCAD commands currently registered
- profile preference state

Workbench-specific commands can appear unavailable until the corresponding native workbench has been activated once. Activate Part Design, Sketcher, Assembly, and TechDraw, then rerun the smoke test if needed.

Manual source-tree installation

This method is intended for developers.

1. Create <FreeCAD user data>/Mod/FusionLikeUI/.
2. Copy packaging/Init.py and packaging/InitGui.py into it.
3. Copy src/fusion_like_ui_runtime.py into it with that exact filename.
4. Copy packaging/installed_README.txt as README.txt.
5. Restart FreeCAD.

The release installer is preferred for testers because it also handles original-state capture, loaded-runtime replacement, version checking, and immediate activation.

First-run setup

1. Start in **DESIGN** and confirm middle-mouse pan, Shift+middle-mouse orbit, and mouse-wheel zoom.
2. Create or edit a Sketch and confirm that the ribbon changes to Sketch tools.
3. Open **DRAWING** once so TechDraw registers its actions.
4. Open **ASSEMBLE** once so the Assembly actions and diagnostic wrappers are registered.
5. Run the smoke test.

Restore without uninstalling

Choose:

Fusion-like → Restore original FreeCAD interface

This disables the profile, removes its runtime widgets and event filters, restores saved navigation preferences, reactivates the original workbench when available, and restores the saved Qt window state.

To re-enable it later, choose:

Fusion-like → Apply / rebuild Fusion-like profile

Uninstall

1. Close active task dialogs.
2. Run macros/Uninstall_FusionLike_FreeCAD_v1.3.0.FCMacro.
3. Restart FreeCAD once.

The uninstaller attempts to restore the original interface before removing the user module.

Installation troubleshooting

Installer reports that FreeCAD is too old

Install FreeCAD 1.1 or newer. The runtime depends on APIs and commands not consistently available in earlier releases.

Nothing appears after installation

Open **View → Panels → Report view** and look for messages prefixed [Fusion-like UI]. Restart FreeCAD once, then rerun the smoke test.

Some toolbar buttons are missing

Activate the corresponding native workbench once. FreeCAD registers some actions lazily.

The interface is duplicated

Choose **Fusion-like → Apply / rebuild Fusion-like profile**. The rebuild removes existing profile toolbars before recreating them.

The original layout is not restored exactly

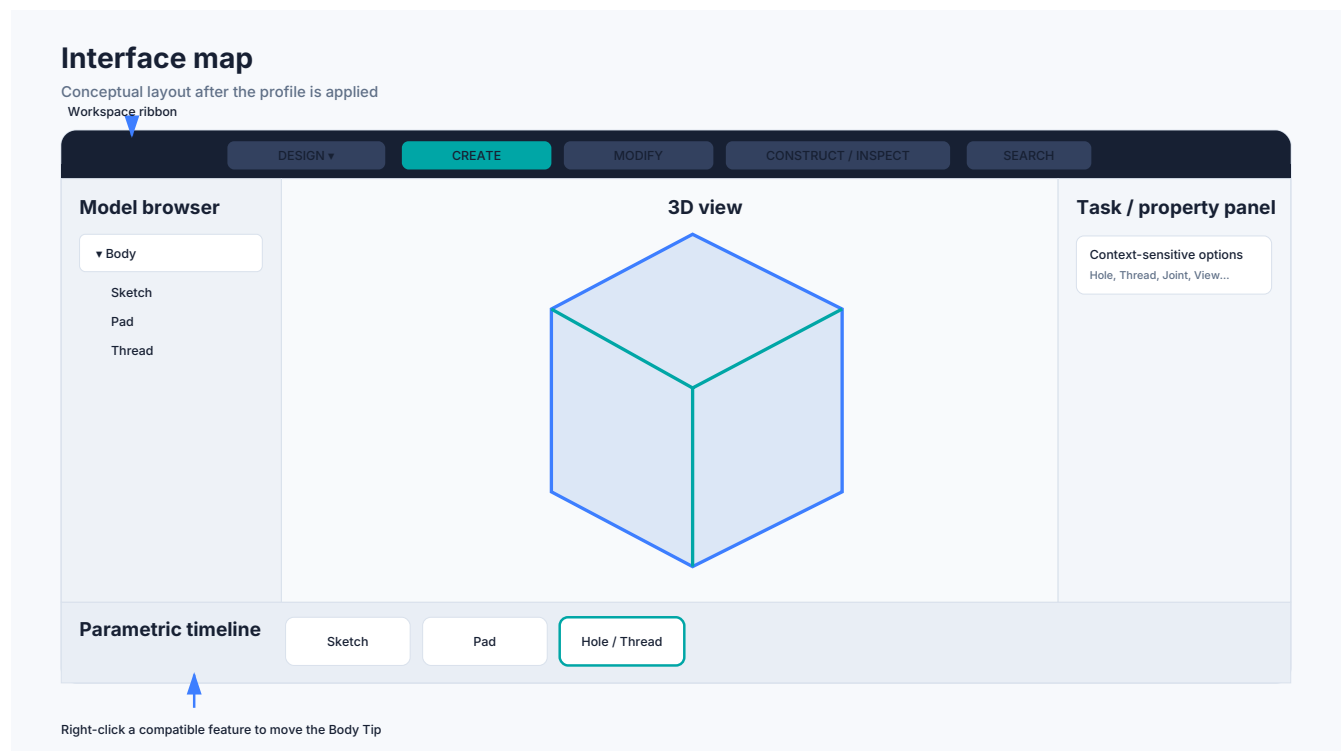
Qt layout state can differ when dock widgets or workbenches have changed between FreeCAD builds. Use FreeCAD's native **View → Panels** and **View → Toolbars** menus to make final adjustments, then report the build and steps used.

Ten-minute quick start

This walkthrough confirms the major workflows without requiring a production model.

1. Confirm the interface

Choose **DESIGN** in the workspace selector. You should see Create, Modify, and Construct/Inspect groups, a model browser on the left, a task/property area, and the feature timeline along the bottom.



2. Create a sketch and solid

1. Create a new document.
2. Create a Part Design Body.
3. Start a Sketch on the XY plane.
4. Confirm that the ribbon changes to Sketch primitives and constraints.
5. Draw a rectangle with **R**.
6. Leave the Sketch.
7. Press **E** and choose **Add / Pad**.

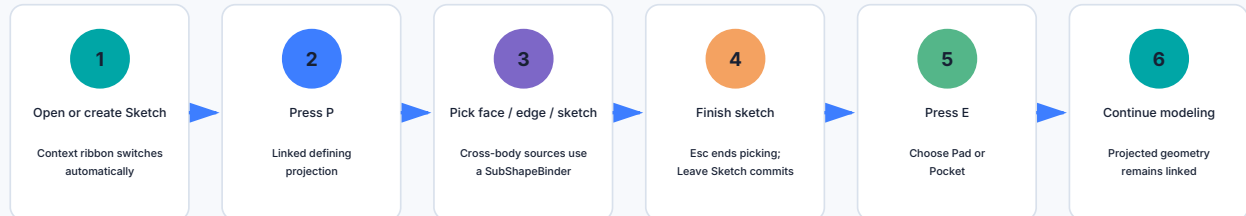
3. Project a face and cut it

1. Select a face of the Pad and create another Sketch on it.
2. Press **P**.

3. Click an outer edge or a suitable face boundary.
4. Press **Esc** to end projection picking.
5. Add any additional sketch geometry required for a closed profile.
6. Press **E** and choose **Cut / Pocket**.

Sketch → Project → Extrude

Persistent references from imported or earlier geometry



Reference modes

P — defining geometry for Pad/Pocket

Shift+P — construction/reference geometry

Intersect — sketch-plane intersection

4. Try a threaded hole

1. Create a Sketch containing a point or circle where the hole should be placed.
2. Select the Sketch.
3. Press **H**.
4. Choose **Hole & Thread**.
5. Select a standard and size.
6. Use cosmetic representation for a fast test, or modeled representation for physical helical geometry.

5. Create a dimensionable drawing

1. Choose **DRAWING**.
2. Click **Insert Model View**.
3. Select the Body as the source and create a new default sheet.
4. Complete the base/projected-view task.
5. On the drawing page, select an actual projected edge or vertex.
6. Use **Smart Dimension**.

Do not use **Active View Snapshot** for this test; it creates a raster image.

Drawing: model view vs. viewport snapshot

Only model-linked TechDraw geometry is normally dimensionable

Correct engineering-drawing path

1. New Sheet

Create or select a TechDraw page.

2. Insert Model View

Select a Body, Part, Link, component, or Assembly.

3. Place base / projected views

Finish the native orientation task.

4. Select drawing edges or vertices

Then use Smart Dimension or a specific dimension tool.



Reference image path

Active View Snapshot

Captures the open 3D viewport as an image.



Raster image — no associative edges for dimensions

6. Create a small assembly

1. Save the document or open two saved component documents.
2. Choose **ASSEMBLE**.
3. Create or activate an Assembly.
4. Select a source Part or Body and click **Add Selected** twice, or duplicate an inserted component with **Ctrl+D**.
5. Ground one component.
6. Select compatible connector geometry on two different instances.
7. Create a native joint.
8. Run **Solve and show diagnostics**.

Assembly: add, duplicate, mate, diagnose

A repeatable component-instance workflow

1 Add Selected

Create App::Link instances in the active Assembly.

2 Copy / Paste

Ctrl+C, Ctrl+V, Ctrl+Shift+V, Ctrl+D.

3 Ground one component

Establish the assembly reference frame.

4 Pick two connectors

Use geometry on two different component instances.

5 Create joint

Fixed, revolute, cylindrical, slider, ball, distance...

6 Read diagnostics

Preflight, live task panel, and decoded solver status.

Mate failure report includes:

connector count · component identity · geometry type · grounding · closed loops · decoded solver code

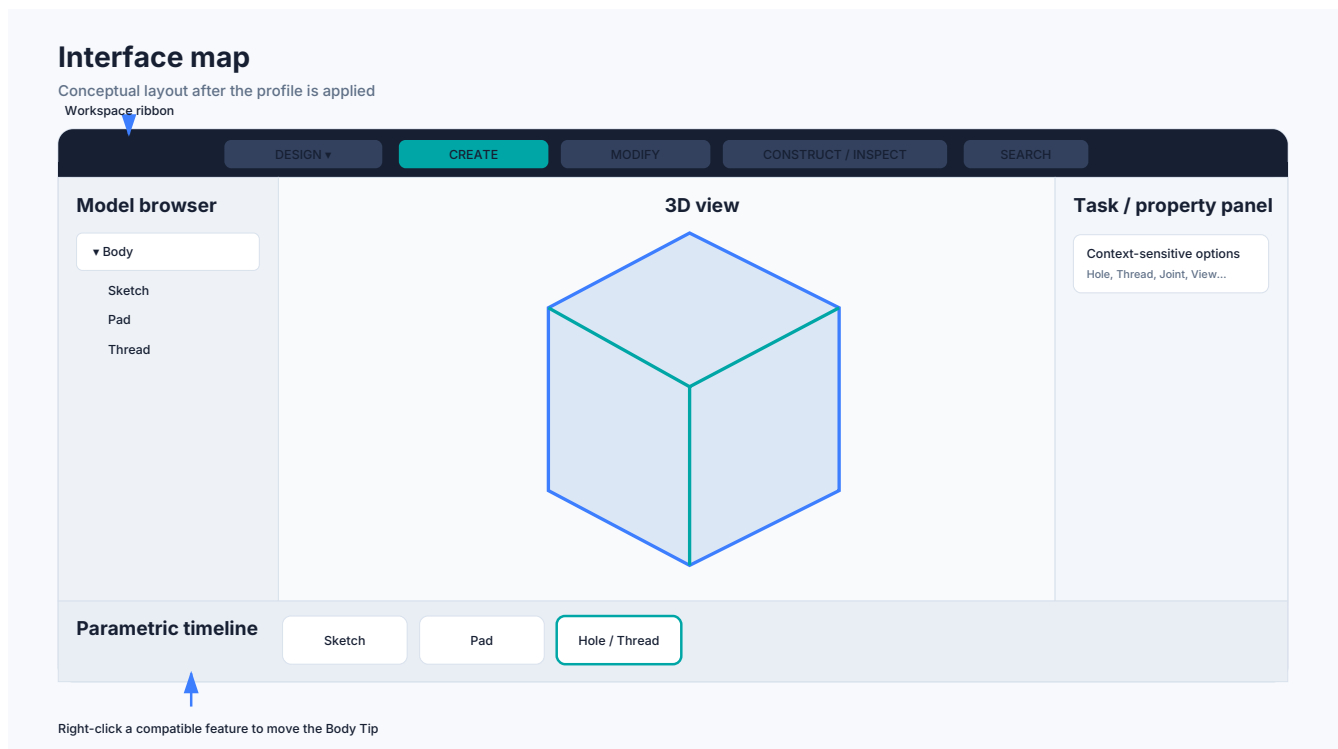
7. Verify recovery

Choose **Fusion-like → Restore original FreeCAD interface**, then reapply the profile. This confirms that the reversible state path works before you begin public testing.

Graphical user guide

This guide describes the tester-facing behavior of version 1.3.0. The illustrations are conceptual diagrams, not screenshots from a specific FreeCAD theme or operating system.

Interface orientation



The top workspace selector determines the main ribbon:

- **DESIGN**: solid Part Design, Sketch entry, projection, Hole, Thread, and inspection.
- **ASSEMBLE**: component insertion, copy/paste, grounding, native joints, motion, BOM, and diagnostics.
- **DRAWING**: TechDraw sheets, model views, projected/section/detail views, dimensions, annotations, symbols, BOM, and export.

Sketch edit mode overrides the workspace ribbon temporarily. Leaving the Sketch restores the previously active workspace ribbon.

Design workspace

Create

- Body and Sketch creation
- Pad/Pocket under **Extrude**
- Hole and threaded-hole workflow
- Revolution/Groove

- additive/subtractive Loft and Sweep/Pipe
- independent Thread

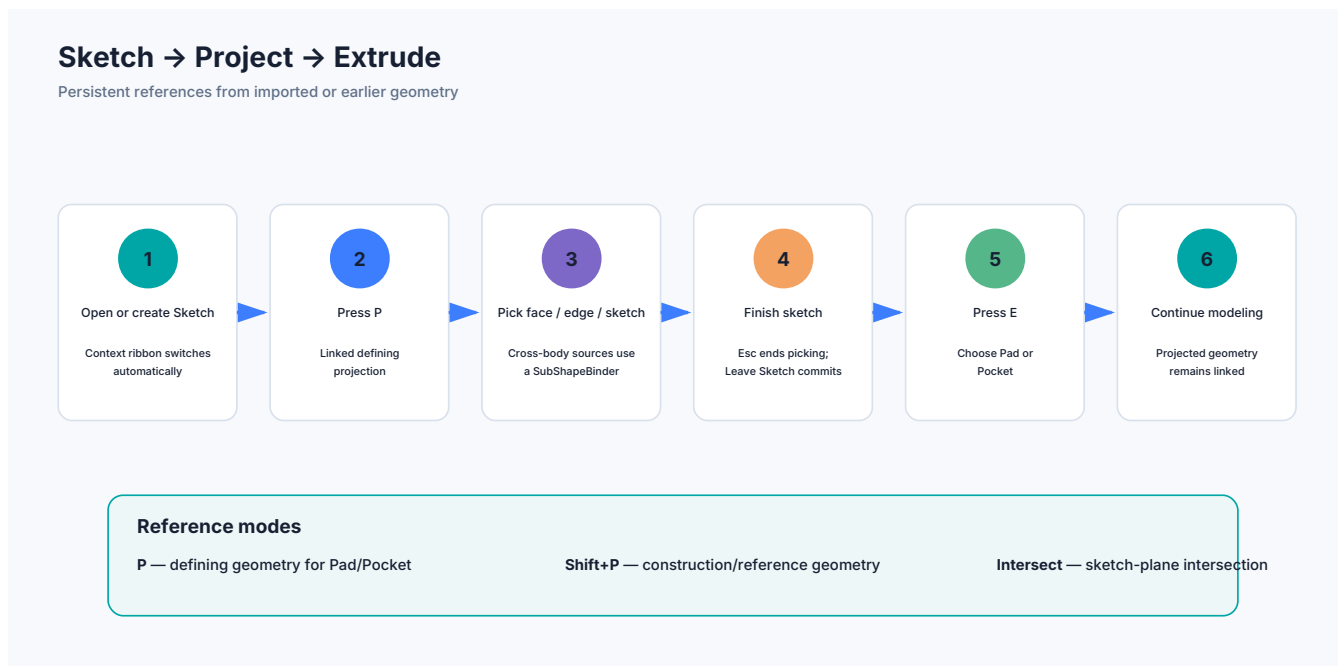
Modify

- Fillet, chamfer, draft, and thickness
- Press/Pull approximation using native Pad, Pocket, Thickness, Draft, and Transform commands
- mirror and pattern

Construct / Inspect

- datum plane, line, point, and coordinate system
- measure, mass properties, geometry check
- fit, isometric, and visibility controls

Sketch and Project / Include



Contextual Sketch ribbon

When a Sketcher::SketchObject enters edit mode, the profile presents:

- **Finish:** Leave Sketch, Cancel Sketch, View Sketch, Section View, Stop operation
- **Create:** point, line/polyline, rectangle, arc, circle/conic, slot, text, construction geometry
- **Project / Include:** defining projection, reference projection, intersection, Carbon Copy
- **Modify:** trim, split, extend, fillet/chamfer, offset, move, rotate, scale, symmetry, clipboard tools
- **Constrain:** dimension and geometric constraints
- **Inspect:** validation and conflict/redundancy selectors

Linked defining projection — P

Use when the projected result should participate in a closed Pad or Pocket profile.

1. Edit the destination Sketch.
2. Press **P**.
3. Select faces, edges, vertices, or an earlier Sketch.
4. Press **Esc** when finished.
5. Complete the profile and use **E** for Pad/Pocket.

Linked construction/reference projection — Shift+P

Use for persistent design references that should not form the solid profile by themselves.

Sketch-plane intersection

Creates linked defining curves where the selected geometry intersects the active sketch plane.

Direct link vs. SubShapeBinder

The runtime links geometry directly when FreeCAD permits the dependency. Imported geometry, cross-Body geometry, and some cross-Part references are routed through a synchronized `PartDesign::SubShapeBinder` inserted before the destination Sketch.

The operation is rejected when it would create a circular dependency, reference a later feature in the same Body, or link directly to another FreeCAD document.

Hole and Thread

Native Hole & Thread — H

The dialog is a Fusion-style front end over a native `PartDesign::Hole`. It supports installed FreeCAD standards, sizes, classes, depth behavior, direction, modeled/cosmetic thread, counterbore/countersink/counterdrill, drill point, taper, and clearance.

Recommended workflow:

1. Select a Sketch containing points, circles, or arcs, or select an existing Hole.
2. Press **H**.
3. Choose **Hole & Thread**.
4. Set hole type, standard, size, class, direction, depth, and representation.
5. Accept and recompute.

Independent cylindrical Thread — Shift+T

Use on one or more cylindrical faces of the current Body Tip. The result is a persistent PartDesign::FeaturePython that can model a helical cut or store cosmetic callout metadata.

- Select cylindrical faces from the same source feature.
- Press **Shift+T**.
- Choose internal/external/automatic recognition, standard, size, class, direction, extent, offset, and clearance.
- Select the Thread feature and press **Shift+T** again to edit it.

Modeled threads increase recompute and file complexity. Cosmetic threads are normally better for large fastener populations.

Drawing workspace

Drawing: model view vs. viewport snapshot

Only model-linked TechDraw geometry is normally dimensionable

Correct engineering-drawing path

1. New Sheet

Create or select a TechDraw page.

2. Insert Model View

Select a Body, Part, Link, component, or Assembly.

3. Place base / projected views

Finish the native orientation task.

4. Select drawing edges or vertices

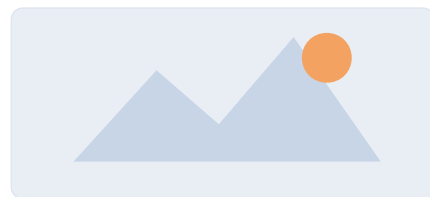
Then use Smart Dimension or a specific dimension tool.



Reference image path

Active View Snapshot

Captures the open 3D viewport as an image.



Raster image — no associative edges for dimensions

Insert Model View

This is the normal engineering-drawing path.

1. Create or select a TechDraw sheet.
2. Click **Insert Model View**.
3. Choose one or more source Bodies, Parts, Links, components, or a native Assembly.
4. Select an existing sheet or create the default sheet.
5. Complete the native base/projected-view task.
6. Select actual drawing edges or vertices on the page.
7. Use Smart Dimension or a specific dimension command.

Active View Snapshot

This inserts a raster image of the open 3D viewport. It is useful for shaded reference illustrations but does not contain associative model edges or vertices and is not normally dimensionable.

Dimension selection

The profile warns when:

- nothing is selected
- only a Page or model-tree view item is selected
- the selection is a raster Active View
- no drawing Edge, Vertex, or Face subelement is selected

Other drawing functions

- projected, section, detail, broken, and clipped views
- hatching and view stacking
- linear, radial, diameter, angular, area, chain, coordinate, and arc dimensions
- centerlines, center marks, thread graphics, leaders, balloons, weld and surface-finish symbols
- thread callouts linked to native Hole or independent Thread features
- Assembly BOM creation and Spreadsheet/BOM placement
- PDF, SVG, and DXF export

Assembly workspace

Assembly: add, duplicate, mate, diagnose

A repeatable component-instance workflow

1 Add Selected

Create App::Link instances in the active Assembly.

2 Copy / Paste

Ctrl+C, Ctrl+V,
Ctrl+Shift+V, Ctrl+D.

3 Ground one component

Establish the assembly reference frame.

4 Pick two connectors

Use geometry on two different component instances.

5 Create joint

Fixed, revolute, cylindrical, slider, ball, distance...

6 Read diagnostics

Preflight, live task panel, and decoded solver status.

Mate failure report includes:

connector count · component identity · geometry type · grounding · closed loops · decoded solver code

Add and duplicate components

- **Add Selected**: insert selected model objects at their current placement
- **Ctrl+C**: copy selected component sources
- **Ctrl+V**: paste new instances with an X offset
- **Ctrl+Shift+V**: paste in place
- **Ctrl+D**: duplicate selected components with an offset

Instances are normally native App: :Link objects. Native subassemblies use Assembly: :AssemblyLink when the build exposes it.

Save both documents before linking components across files. The profile rejects dependency loops and reports skipped sources.

Grounding

Ground at least one component or establish a grounded path. A fully floating assembly has no reference frame and returns the native solver status for “no grounded component.”

Connector selection

Choose geometry on two different component instances:

Joint type	Good connector choices
Fixed	faces, edges, vertices, datums, axes, coordinate systems
Revolute / cylindrical	cylindrical faces, circular edges, explicit axes
Slider	axes, straight edges, planar references, coordinate systems
Ball	vertices, centers, point-like references
Parallel / perpendicular / angle	planar faces, straight edges, axes, datum planes
Distance	point, edge, face, axis, or datum references appropriate to the relation

Diagnostic stages

1. **Preflight** checks connector count, component identity, subelement type, obvious geometry mismatch, grounding, and likely closed loops.
2. **Live task panel** updates while the native joint task is open.
3. **Post-solve report** decodes the native solver result and selects a newly failed joint when possible.

Native solver results decoded by the profile:

Code	Meaning
0	solved
-6	no grounded component
-4	over-constrained
-3	conflicting joints
-5	malformed/broken joint
-1	general solver error
-2	redundant joints

Timeline and command search

Parametric timeline

The bottom timeline visualizes compatible document objects in document order. Right-click a compatible Part Design feature to set the Body Tip there or restore the latest solid feature. This is a Body-Tip workflow aid, not Fusion’s native timeline engine.

Command search — S

Searches registered FreeCAD QAction commands and triggers the selected action. Disabled commands are shown but cannot be run.

Keyboard map

Keyboard map

Context-sensitive shortcuts; text inputs and modal dialogs are protected

S

Command search

E

Pad / Pocket

Q

Press / Pull

H

Hole / Thread

Shift+T

Independent Thread

P

Project profile

Shift+P

Project reference

Esc

End projection

F

Fillet

M

Move / Transform

V

Visibility

I

Measure

L

Sketch line

R

Rectangle

C

Circle

T

Trim

O

Offset

D

Dimension

X

Construction

Ctrl+C

Copy component

Ctrl+V

Paste offset

Ctrl+Shift+V

Paste in place

Ctrl+D

Duplicate component

Shortcuts are ignored while focus is in a text field, spin box, combo box, modal dialog, or unrelated window.

Restore and recovery

Use **Fusion-like → Restore original FreeCAD interface** to remove profile widgets and restore saved layout/navigation state. Use **Apply / rebuild** to re-enable it.

For errors, open **View → Panels → Report view** and capture all relevant lines prefixed [Fusion-like UI].

Troubleshooting

First diagnostic steps

1. Open **View** → **Panels** → **Report view**.
2. Reproduce the problem once.
3. Copy all lines prefixed [Fusion-like UI] plus any adjacent Python traceback.
4. Run the v1.3 smoke-test macro.
5. Record the FreeCAD version from **Help** → **About FreeCAD** → **Copy to clipboard**.

Sketch ribbon does not appear

- Confirm that a Sketcher::SketchObject is actually in edit mode.
- Wait a fraction of a second for late Sketcher command registration.
- Choose **Fusion-like** → **Apply / rebuild Fusion-like profile**.
- Activate the Sketcher workbench once, then return to Part Design and edit the Sketch again.
- Check Report view for `_refresh_ui_context_if_needed` errors.

Sketch primitive button is missing

The runtime uses grouped-command IDs first and individual-command fallbacks. A missing button can indicate that the corresponding native command is not registered in that build. Search for the primitive with **S** and include its action object name in the issue report.

Projection fails

Common causes:

- destination Sketch is not in edit mode
- source is the destination Sketch itself
- source feature occurs later in the same Body
- source already depends on the destination Sketch
- cross-document direct links are attempted
- the selected subelement was deleted or renamed by a topology change

For cross-document workflows, import or link the source into the destination document first.

Pad or Pocket cannot use a projected profile

- Confirm that **P**, not Shift+P, created defining geometry.
- Check that the resulting loops are closed and non-self-intersecting.
- Confirm the Sketch belongs to the active Body.

- Use Sketch validation and inspect redundant/conflicting constraints.

Modeled thread is slow or fails

- Try cosmetic mode to verify the standard/size metadata first.
- Reduce thread length.
- Remove extreme custom clearance.
- Confirm the selected face is cylindrical and belongs to the current Body Tip source.
- Check for self-intersecting helical profiles or a boolean failure in Report view.

Drawing view cannot be dimensioned

- Confirm it is a model-linked TechDraw view, not ActiveView / DrawViewImage.
- Select the actual projected edge or vertex on the drawing page.
- Wait for HLR recomputation to finish.
- Confirm the source object has a valid shape.
- Try a specific dimension command after Smart Dimension explains the selection.

Assembly component cannot be added

- Activate or create an Assembly.
- Save both source and Assembly documents for cross-document links.
- Confirm the source object has a shape or is a supported Part/Body/Link.
- Avoid linking a document back into itself through a dependency cycle.

Two parts look compatible but will not mate

Check in this order:

1. The references belong to two different **component instances**, not two faces of the same source object.
2. One component is grounded or connected to ground.
3. Connector geometry matches the joint type: axial references for revolute/cylindrical; point-like references for ball.
4. An earlier joint has not already removed the same degrees of freedom.
5. Offsets and reverse direction are not forcing an impossible placement.
6. Linked face/edge names are still valid after source-model edits.

Run **Why won't this mate?** before creation and **Solve and show diagnostics** afterward.

Solver status guide

- -6: ground a component
- -4: suppress the newest joint or replace multiple constraints with a single intended joint
- -3: inspect conflicting joints and connector directions
- -5: re-pick broken references

- -2: remove redundant joints
- -1: simplify the graph, check offsets, and report a minimal case

Restore does not recover the exact window layout

Qt window-state serialization depends on available docks and workbenches. Manually restore remaining panels through **View → Panels** and toolbars through **View → Toolbars**, then report the FreeCAD build and before/after state.

Known limitations

Not an implementation clone

The project reorganizes and extends FreeCAD workflows. It does not reproduce Autodesk's kernels, cloud data model, T-spline environment, timeline engine, assembly solver, CAM system, or drawing implementation.

Topological naming

References to FaceN, EdgeN, and VertexN can break when upstream topology changes. FreeCAD's element naming helps but cannot guarantee every reference survives a major feature rewrite. Projection, threads, dimensions, and joints may need repair.

Independent Thread dependency

The independent Thread is a custom `PartDesign::FeaturePython`. Its last saved shape can remain visible without the profile, but editing or recomputing it requires the runtime to be installed.

Modeled thread cost

Physical helical threads can greatly increase recomputation time, file size, boolean complexity, meshing time, and drawing HLR time. Use cosmetic threads unless physical flank geometry is required.

Thread standards

The independent cylindrical Thread intentionally omits standalone NPT-style tapered-face behavior. Use the native threaded Hole workflow for tapered internal threads supported by FreeCAD.

Projection boundaries

Direct cross-document Sketch external geometry is not supported. Cross-Body and imported geometry are mediated through `SubShapeBinders` when valid. Circular and future-history dependencies are rejected.

Drawing snapshots

Active View Snapshot is a raster image by design. It cannot be converted automatically into an associative model view and is not a source for normal TechDraw dimensions.

Drawing dimensions

The profile validates common selection mistakes but does not replace TechDraw's dimension engine. Dimension semantics, HLR completion, and reference repair remain native FreeCAD behavior.

Assembly diagnostics

Preflight is advisory. Assembly compatibility depends on the complete nonlinear constraint graph, connector frames, offsets, grounding, and solver state. Geometrically similar parts are not guaranteed to solve under a chosen joint type.

Assembly document links

Cross-document components require saved files and stable source object identities. Moving, renaming, closing, or replacing source documents can break links.

UI themes and platform differences

Qt widget sizes, action registration timing, menu text, and icons vary across platforms and FreeCAD builds. The profile uses command IDs and compatibility fallbacks, but some visual differences are expected.

Static validation boundary

Repository CI can compile and inspect the code without importing FreeCAD. It cannot exercise the live GUI, FreeCAD document model, Qt signal behavior, or Open CASCADE geometry operations.

Public testing guide

Test philosophy

The repository supports three levels of validation:

1. **Static validation** outside FreeCAD: syntax, embedded payload equality, internal method resolution, documentation links, release contents, and ZIP integrity.
2. **Smoke testing** inside FreeCAD: installation state and command registration without modifying a document.
3. **Manual workflow testing** inside FreeCAD: real GUI interactions and document recomputation.

Static checks cannot prove GUI or Open CASCADE behavior. Public test reports are essential.

Static validation

From the repository root:

```
python tools/validate_package.py --repo .  
python tools/build_release.py --repo . --check
```

The GitHub Actions workflow runs the same checks on pull requests.

FreeCAD smoke test

Install the profile, then execute:

```
macros/FusionLikeUI_SmokeTest_v1.3.0.FCMacro
```

Record its output from Report view.

Manual test matrix

Installation and recovery

- **I-01** Fresh install into a clean FreeCAD user profile.
- **I-02** Upgrade directly from v1.0, v1.1, and v1.2 where available.
- **I-03** Rebuild the profile twice; confirm no duplicate toolbars.
- **I-04** Restore original interface, re-enable, then uninstall.
- **I-05** Restart FreeCAD and confirm automatic startup.

Sketch context

- **S-01** Enter Sketch from Part Design; confirm contextual ribbon appears.
- **S-02** Leave Sketch; confirm Design ribbon returns.
- **S-03** Test line, rectangle, circle, arc, slot, trim, offset, dimension, and construction commands.
- **S-04** Open Sketch immediately after workbench activation; confirm delayed command registration triggers a rebuild.

Projection

- **P-01** Project a same-Body earlier edge and Pad/Pocket the result.
- **P-02** Project an entire planar face boundary.
- **P-03** Project a whole earlier Sketch.
- **P-04** Project imported STEP geometry through a binder.
- **P-05** Project across Bodies.
- **P-06** Confirm later-feature and circular-reference attempts are rejected.
- **P-07** Change the source dimensions and confirm linked projection updates.

Hole and Thread

- **H-01** Native plain, counterbore, and countersink holes.
- **H-02** Cosmetic and modeled ISO metric threads.
- **H-03** Left-hand and partial-depth thread.
- **T-01** External independent thread on a cylinder.
- **T-02** Internal independent thread where supported by selected face geometry.
- **T-03** Edit an existing independent Thread with Shift+T.
- **T-04** Save, close, reopen, and recompute.

Drawing

- **D-01** Insert Model View from a Body and add projected views.
- **D-02** Dimension visible linear, circular, and angular geometry.
- **D-03** Confirm Active View Snapshot is identified as raster.
- **D-04** Insert an Assembly as a source.
- **D-05** Create section/detail views and hatching.
- **D-06** Create a thread callout.
- **D-07** Export PDF, SVG, and DXF.

Assembly

- **A-01** Add Selected from same document.
- **A-02** Add a component from a saved external document.
- **A-03** Copy, paste with offset, paste in place, and duplicate.
- **A-04** Ground a component and create fixed/revolute/cylindrical/slider/ball joints.
- **A-05** Deliberately select two connectors from the same component; verify diagnostic.
- **A-06** Deliberately create a redundant or conflicting joint; verify decoded result.
- **A-07** Test broken linked subelements after changing source topology.
- **A-08** Create exploded view, BOM, and simulation where applicable.

Recommended test environments

Report the exact environment rather than assuming equivalence:

- FreeCAD version and build hash
- Windows, macOS, or Linux distribution
- Qt/PySide generation if known
- Open CASCADE version if shown by About FreeCAD
- graphics backend/driver for display-specific failures

Bug report minimum

A useful report includes:

1. test ID or exact workflow
2. minimal .FCStd file when shareable
3. starting selection and active workbench
4. expected result
5. actual result
6. all [Fusion-like UI] messages and traceback
7. whether the problem persists after restart and profile rebuild

Remove confidential geometry before uploading a document.